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Bwrdd Iechyd Prifysgol
Betsi Cadwaladr
University Health Board

ANAESTHESIA, CRITICAL CARE & PAIN MANAGEMENT DIRECTORATE

OBSTETRIC ANAESTHESIA WRITTEN CONTROL DOCUMENT

MANAGEMENT OF ACCIDENTAL DURAL PUNCTURE

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Author(s):	Dr Gareth Roberts	**Author(s) title:**	Consultant Anaesthetist
Responsible dept / director:	Department of Anaesthesia, BCUHB (West) Dr Linda Warnock Clinical Lead of Anaesthesia, BCUHB (West)		
Approved by:	Department of Anaesthesia, BCUHB (West)		
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PROPRIETARY INFORMATION

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1. Introduction and Purpose

Accidental dural puncture is a recognised complication of epidural anaesthesia, occurring in around 1% of epidural insertions. Correct management of the patient is essential to ensure adequate analgesia during labour and to reduce the risk of causing further dural punctures.

2. Scope

This guideline is intended as a reference for trainee anaesthetists seeking advice regarding the management of accidental dural puncture.

3. Points to consider

3.1 Recognition

- Obvious free flow of clear straw coloured fluid from Tuohy needle on removal of LOR syringe
- Straw coloured fluid freely aspirated using syringe via Tuohy needle
- Straw coloured fluid freely aspirated from an epidural catheter in situ
- Sensory block to T4 - T6 along with a dense motor block, within 5 minutes of injecting test dose of 15mls 0.1 % levobupivacaine + 2 micrograms / ml fentanyl via epidural catheter
- If in doubt, then confirm that fluid is CSF using dextrostix (CSF will be positive for glucose)

3.2 Procedure

- Inform duty Consultant Anaesthetist immediately
- If Tuohy needle is still in situ when the accidental dural puncture diagnosis is made then thread epidural catheter intrathecally leaving 3 – 4 cm of catheter in the intrathecal space. Fix catheter to the skin using clip, Tegaderm and Mefix dressings.

  THE CATHETER MUST BE LABELLED AS A SPINAL CATHETER to avoid accidental injection of large (epidural) doses of local anaesthetic
- Explain to the patient what has happened and that she probably now has a 70% risk of developing a post dural puncture headache
- Give 1,000 mls Hartmanns solution over 30 minutes and be prepared to manage hypotension

- Monitor CTG

- Allow spinal block to recede down to T10 or until patient starts complaining of pain

- **ANAESTHETIST ONLY** then to give top ups of 1 ml 0.5% plain levobupivacaine until adequate analgesia achieved with sensory block height < T6. Top ups will usually need to be given at roughly hourly intervals until delivery. Remember to flush 1ml saline through the epidural filter and line after the top up, to account for the dead space in these areas.

- Avoid intrathecal opiates

- The event does not provide an indication for elective instrumental delivery

- If patient requires a Caesarean Section then inform duty Consultant Anaesthetist and proceed to top up spinal block using 1 ml increments of 0.5% bupivacaine until sensory level of block reaches T4.Wait 5 – 10 minutes in between each 1ml 0.5% bupivacaine top up to allow the LA to take effect

- If accidental dural puncture diagnosis is made once catheter has been sited in subarachnoid space by aspiration of CSF from the catheter then re-label catheter as a spinal catheter

- If accidental dural puncture diagnosis has been made, but the Tuohy needle tip has already been withdrawn from the subarachnoid space, then following discussion with duty Consultant Anaesthetist offer the patient the option of having the epidural re-sited at a different lumbar interspace by the Consultant himself. In such patients an epidural infusion can be used, but start the infusion at 10ml/hour and ensure careful monitoring of sensory block height and motor block. Alternatively convert to conventional intramuscular opiate analgesia or fentanyl PCA plus Entonox

3.3 <u>Post delivery management</u>

- Remove the spinal / epidural catheter carefully using an aseptic technique

- Midwive should alert the anaesthetist if the patient develops a post dural puncture headache (PDPH)

- Patient should have at least a daily follow up by the anaesthetist post delivery

- If a post dural puncture headache develops then :

  a) Inform duty Consultant Anaesthetist
  b) Conservative treatment for 24 hours including :
     i. Bed rest for 24 hours

     ii. Avoid straining and lifting

iii. Ensure adequate oral (or alternatively intravenous) fluid intake (2.5 – 3 litres / 24 hours) especially caffeine containing drinks
iv. Simple analgesics eg. Paracetamol or NSAIDS eg. sodium diclofenac (avoid codeine containing analgesics if breastfeeding)
v. Ensure thromboprophylaxis using Enoxaparin (Clexane) 40mg subcutaneously once daily and TED leg stockings or mechanical calf compression pump
vi. Daily review of patient by anaesthetist
vii. Provide a copy of the OAA Patient Information leaflet on PDPH

- In the case of a PDPH not resolving following 24 hours of conservative treatment or the patient developing deafness or other cranial nerve palsy, the patient should be offered an epidural blood patch (see Epidural Blood Patch Guidelines)

- On discharge of patient from hospital :

i. Complete a discharge letter for the GP
(use the GP letter template available in PDPH drawer on Llifon ward & provide a copy of the OAA Patient Information Leaflet on PDPH with the discharge letter)

ii. Notify Dr G. Roberts Lead Obstetric Anaesthetist
(BCUHB – West) of the PDPH. Please provide details including patient addresograph, date of onset of PDPH and details of the clinical management . He will then arrange an OPD follow up within 2 – 3 weeks of hospital discharge.

4. References

1. Baraz R, Collis RE. The management of accidental dural puncture during labour epidural analgesia: a survey of UK practice *Anaesthesia* 2005; **60**: 673-679